

$$1. y = f(3x^2 - 2x) \quad y' = f'(3x^2 - 2x)[6x - 2]$$

$$2. y = g(\sin^4 x) \quad y' = g'(\sin^4 x) \cos 4x (4) \\ = 4g'(\sin^4 x) \cos 4x$$

$$3. y = \cos(f(x^2 + 1)) \quad y' = -\sin(f(x^2 + 1))(f'(x^2 + 1))(2x) \\ y' = -2x \sin(f(x^2 + 1)) f'(x^2 + 1)$$

$$4. y = f(g(2x^3 - x)) \quad y' = f'(g(2x^3 - x)) g'(2x^3 - x) [6x^2 - 1]$$

$$5. y = \sin(g(\cos(3x))) \quad y' = \cos(g(\cos(3x))) g'(\cos(3x)) [-\sin(3x)] (3) \\ = -3 \cos(g(\cos(3x))) g'(\cos(3x)) \sin(3x)$$

$$6. y = g(\cos(f(2x^2 - 5x))) \quad y' = g'(\cos(f(2x^2 - 5x))) (-\sin(f(2x^2 - 5x)) f'(2x^2 - 5x) (4x - 5)) \\ = -g'(\cos(f(2x^2 - 5x))) \sin(f(2x^2 - 5x)) f'(2x^2 - 5x) [4x - 5]$$

$$7. y = f(\sin(g(x^2 - 4x))) \quad y' = f'(\sin(g(x^2 - 4x))) \cos(g(x^2 - 4x)) g'(x^2 - 4x) [2x - 4]$$

$$8. y = \tan(f(\sqrt{x^2 + 4})) \quad y' = \frac{\sec^2(f(\sqrt{x^2 + 4})) f'(\sqrt{x^2 + 4}) (2x)}{2\sqrt{x^2 + 4}} = \frac{x \sec^2(f(\sqrt{x^2 + 4})) f'(\sqrt{x^2 + 4})}{\sqrt{x^2 + 4}}$$



$$9. y = g(\sin(f(\cos 2x))) \quad y' = g'(\sin(f(\cos 2x))) \cos(f(\cos 2x)) f'(\cos 2x) [-2 \sin 2x]$$

$$10. y = f(\cos(g(\sqrt{x^2 + \sin x}))) \quad y' = \frac{f'(\cos(g(\sqrt{x^2 + \sin x}))) [-\sin(g(\sqrt{x^2 + \sin x})) g'(\sqrt{x^2 + \sin x}) (2x + \cos x)]}{2\sqrt{x^2 + \sin x}}$$

$$11. y = \sin(g(\tan(f(x^3 - \cos x)))) \quad y' = \cos(g(\tan(f(x^3 - \cos x)))) g'(\tan(f(x^3 - \cos x))) \sec^2(f(x^3 - \cos x)) f'(x^3 - \cos x) [3x^2 + \sin x]$$

$$12. y = g(\cos(f(\sin \sqrt{x^2 + 3x}))) \quad y' = \frac{g'(\cos(f(\sin \sqrt{x^2 + 3x}))) [-\sin(f(\sin \sqrt{x^2 + 3x})) f'(\sin \sqrt{x^2 + 3x}) (\cos \sqrt{x^2 + 3x}) (2x + 3)]}{2\sqrt{x^2 + 3x}}$$

$$13. y = f(\sin(g(\cos \sqrt{x^4 - \tan x^2})))$$

$$y' = \frac{f'(\sin(g(\cos \sqrt{x^4 - \tan x^2}))) \cos(g(\cos \sqrt{x^4 - \tan x^2})) g'(\cos \sqrt{x^4 - \tan x^2}) [-\sin \sqrt{x^4 - \tan x^2}] (4x^3 - 2x \sec^2 x^2)}{2\sqrt{x^4 - \tan x^2}}$$

$$= \frac{-f'(\sin(g(\cos \sqrt{x^4 - \tan x^2}))) \cos(g(\cos \sqrt{x^4 - \tan x^2})) g'(\cos \sqrt{x^4 - \tan x^2}) (\sin \sqrt{x^4 - \tan x^2}) (2x^3 - x \sec^2 x^2)}{\sqrt{x^4 - \tan x^2}}$$

$$14. y = g(\tan(f(\sin \sqrt{x^3 - \cos 2x^2})))$$

$$y' = \frac{g'(\tan(f(\sin \sqrt{x^3 - \cos 2x^2}))) \sec^2(f(\sin \sqrt{x^3 - \cos 2x^2})) f'(\sin \sqrt{x^3 - \cos 2x^2}) (\cos \sqrt{x^3 - \cos 2x^2}) (3x^2 + 4x \sin 2x^2)}{2\sqrt{x^3 - \cos 2x^2}}$$

$$15. y = \cos(\sin(g(\tan \sqrt{x^2 + f(x^3)})))$$

$$y' = \frac{-\sin(\sin(g(\tan \sqrt{x^2 + f(x^3)}))) \cos(g(\tan \sqrt{x^2 + f(x^3)})) g'(\tan \sqrt{x^2 + f(x^3)}) \sec^2(\sqrt{x^2 + f(x^3)}) (2x + 3x^2 f'(x^3))}{2\sqrt{x^2 + f(x^3)}}$$